MINI-ERP PROJECT

INVENTORY MANAGEMENT SYSTEM

System Scope, Database Design, Business Rules, and Integration Requirements

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1. PURPOSE AND SYSTEM SCOPE

1.1 System Overview

The Mini-ERP project divides the operations of an online marketplace into five connected systems: Product/Catalog Management, Sales/Order Management, Inventory Management, Billing/Payment Management, and Delivery/Shipment Management. A single customer purchase passes through all five, so each system must know exactly which records it controls and which records it only reads from the others.

The Inventory Management System is the point of truth for stock. Whenever another system needs to know whether a seller can still sell a product, or needs units set aside for a customer, the request comes to Inventory. Inventory answers using its own counts and records every change it makes, so that each stock figure can be traced to the events that produced it.

This document describes the responsibilities, logical database structure, business rules, and integration requirements proposed by the Inventory Management group. The design is written without depending on a particular database platform, since the class has not yet chosen between Firebase and Supabase. Items that affect other groups are listed in Section 6 for joint confirmation.

1.2 Purpose of the System

The main purpose of the Inventory Management System is to keep stock figures accurate while orders are being placed, paid for, and shipped. It does this by separating the units a seller physically holds from the units already promised to customers. The difference between the two is the quantity that can still be sold.

The system also protects the marketplace from overselling. When two customers attempt to buy the last available unit at the same moment, only one reservation may succeed. Reservation, release, and deduction are therefore handled only by Inventory and are processed one at a time for each stock record.

Finally, the system keeps a permanent movement log. Restocks, sales, returns, and corrections each create a log entry that records the quantity, the reason, and the resulting balance. This allows sellers and authorized staff to explain any stock figure and to detect errors without editing past records.

1.3 Functional Scope

The system covers the full life of a stock unit from the time a seller adds it until it leaves with a shipment or returns to the shelf. It responds to reservation requests from Sales, adjusts reservations as orders progress, deducts stock when Delivery reports dispatch, and adds stock back when a returned package is received.

Sellers and authorized staff may record new stock and corrections. The other systems may read availability and reservation results, but they may not change stock figures directly. Table 1 summarizes the functions covered by the system.

Table 1. Functional Scope of the Inventory Management System

Function	Description
Stock record setup	Create one stock record for each seller offer supplied by Product/Catalog Management.
Restocking	Record units added by the seller and increase the quantity on hand.
Availability checking	Report the quantity that can still be sold for a given seller offer.
Stock reservation	Hold units for submitted order items so that other customers cannot buy them.
Reservation confirmation	Remove the expiry from a reservation once Sales reports verified payment.
Reservation release and expiry	Return held units to available stock when an order is cancelled, fails, or is left unpaid.

Function	Description
Stock deduction	Reduce the quantity on hand when Delivery reports that the items were dispatched.
Return receiving	Add units back when a returned package reaches the seller.
Stock adjustment	Record corrections for damaged, lost, or miscounted units with a required reason.
Movement history	Keep a permanent log of every change to the quantity on hand.
Low-stock monitoring	Identify stock records at or below their reorder level.

1.4 Responsibility Boundaries

Inventory owns the stock records, reservations, and movement log. It does not store product descriptions, prices, order contents, payments, or shipment details. Where it needs to connect with those records, it keeps only the identifier used by the owning system. For example, a reservation stores the order item identifier from Sales, but the ordered quantity and price shown to the customer remain in the Sales record.

The same rule works in reverse. Sales may ask Inventory to reserve three units, but only Inventory decides whether the request succeeds and changes the stock figures. Delivery may report that a package left the seller, but only Inventory deducts the units and writes the movement entry. Table 2 shows how record ownership is divided among the five systems.

Table 2. Record Responsibilities of the Mini-ERP Systems

System	Records it is responsible for
Product/Catalog Management	Categories, products, sellers, seller offers, selling prices, and offer status

System	Records it is responsible for
Sales/Order Management	Customers, orders, order items, accepted prices, order totals, and order status
Inventory Management	Stock records, quantities on hand and reserved, reservations, reorder levels, and movement history
Billing/Payment Management	Invoices, payment attempts, payment verification, and refunds
Delivery/Shipment Management	Shipments, courier assignment, tracking, dispatch, and delivery confirmation

1.5 Proposed Initial Demonstration Scope

For the first demonstration, each seller offer has a single stock record and no separate warehouse locations. Stock is counted in whole units. A reservation request for an order is processed as one unit of work: either every item is reserved or none of them is, so the system never leaves a partial reservation behind.

Unpaid reservations are proposed to expire after 30 minutes. Stock is deducted at dispatch rather than at payment, so that a cancellation before dispatch only needs a release. Supplier purchasing, multiple warehouses, batch or expiry-date tracking, and stock transfers between sellers are outside the initial scope. These limits are proposals and must be confirmed with the affected groups.

2. DATABASE DESIGN AND ENTITY RELATIONSHIPS

2.1 Design Overview

The design separates three concerns that change at different rates. The Inventory entity holds the current balance for one seller offer. The Stock Reservation entity records each temporary hold placed for an order item. The Stock Movement entity records every

permanent change to the balance. Keeping these apart allows the current figures to be read quickly while the full history stays intact.

Only two quantities are stored on the stock record: the quantity on hand and the quantity reserved. The quantity available for sale is always computed from them. Storing a third figure would create the possibility of the three numbers disagreeing.

Seller Offer and Order Item appear in the diagram as external entities. Product/Catalog Management owns Seller Offer, and Sales/Order Management owns Order Item. Inventory refers to them by identifier and does not duplicate their data.

2.2 Entity-Relationship Diagram

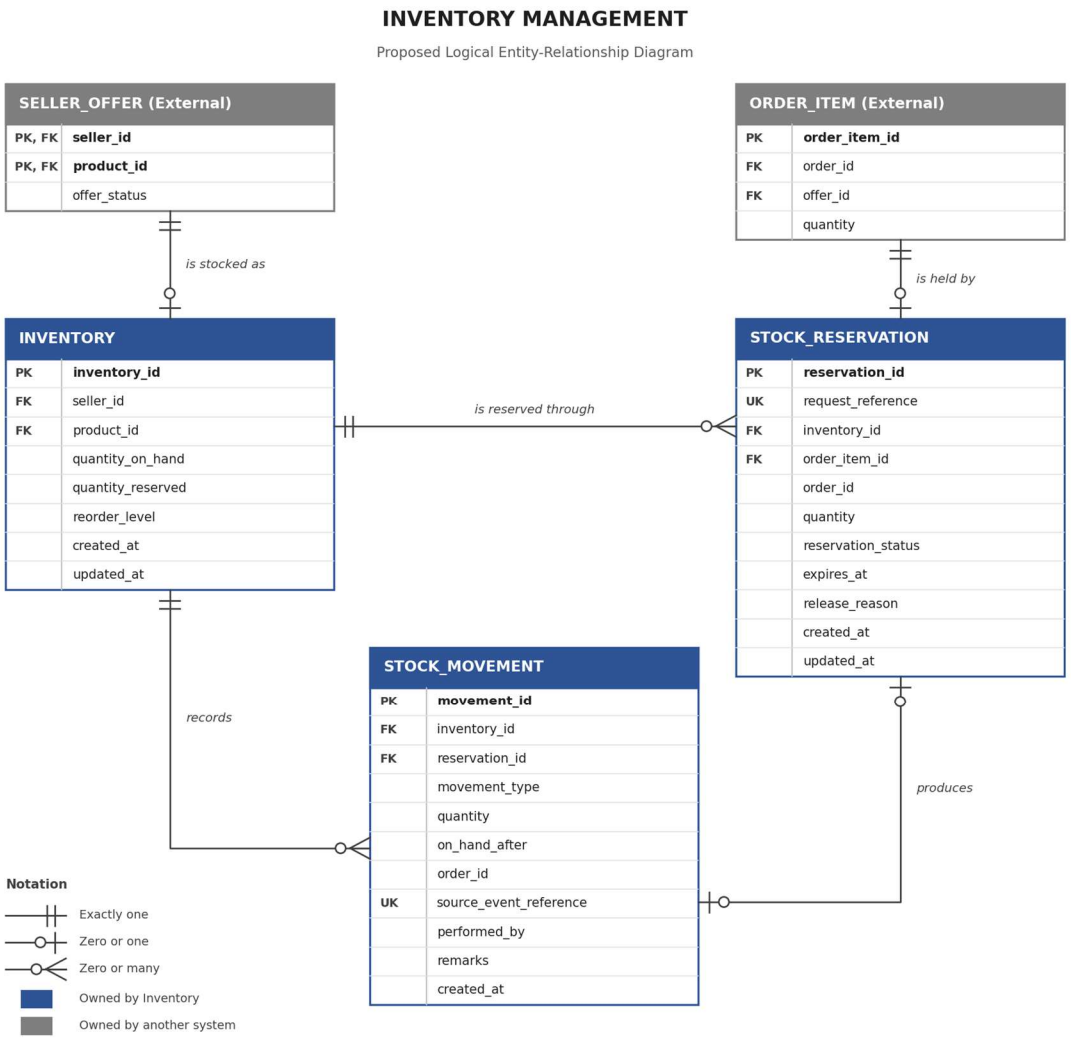


Figure 1. Proposed Logical Entity-Relationship Diagram of the Inventory Management System.

Figure 1 shows that each seller offer may have one stock record, and that each stock record may have many reservations and many movement entries. A reservation always belongs to a single order item, and an order item is held by at most one reservation.

The link between Stock Reservation and Stock Movement is optional on both sides. Restocks, adjustments, and returns produce movements without a reservation, while a

reservation that is released or expires produces no movement at all. Only a fulfilled reservation creates a matching sale movement.

2.3 Entity and Attribute Definitions

The tables below list the proposed attributes. Primary keys identify each record, foreign keys link records to one another, and unique keys prevent the same request or event from being stored twice. If a document database is selected, the same references and checks must be enforced by validation rules. Tables 3 to 7 describe each entity in turn.

2.3.1 Inventory

The Inventory entity stores the current stock balance of one seller offer. It is created when a seller first enters stock for an offer and is updated whenever units are reserved, released, deducted, or added.

The reorder level is set by the seller. When the available quantity falls to or below this figure, the record is reported as low stock so that the seller can restock before the offer runs out.

Table 3. Attributes of the Inventory Entity

Attribute	Key or Constraint	Description
inventory_id	PK	Unique stock record identifier
seller_id	FK, required	Seller who holds the stock
product_id	FK, required	Product being stocked
quantity_on_hand	Required, integer ≥ 0	Units physically held by the seller
quantity_reserved	Required, integer ≥ 0, not greater than quantity_on_hand	Units held for orders that have not been dispatched
reorder_level	Required, integer ≥ 0, default 0	Available quantity at which the record is reported as low stock

Attribute	Key or Constraint	Description
created_at	Required	Date and time the record was created
updated_at	Required	Date and time either quantity last changed

The pair seller_id and product_id must be unique and together reference one Seller Offer, so an offer cannot have two stock records.

2.3.2 Stock Reservation

The Stock Reservation entity records units set aside for one order item. It begins as Active with an expiry time, becomes Confirmed when payment is verified, and ends as Fulfilled, Released, or Expired.

The request reference is supplied by Sales with each reservation request. If the same request arrives again because of a retry or a lost response, Inventory finds the existing reservation through this reference and returns the original result instead of reserving the units a second time. The order identifier is stored alongside the order item so that all holds for an order can be released together.

Table 4. Attributes of the Stock Reservation Entity

Attribute	Key or Constraint	Description
reservation_id	PK	Unique reservation identifier
request_reference	Unique, required	Reference of the Sales request that created the hold
inventory_id	FK, required	Stock record the units are taken from
order_item_id	FK, unique, required	Order item the hold is for
order_id	Required	Order that contains the item

Attribute	Key or Constraint	Description
quantity	Required, integer > 0	Units held
reservation_status	Required	Active, Confirmed, Fulfilled, Released, or Expired
expires_at	Required while Active	Time an unpaid hold is released automatically
release_reason	Required when Released	Cause of the release, such as cancellation or failed payment
created_at	Required	Date and time the hold was placed
updated_at	Required	Date and time of the last status change

2.3.3 Stock Movement

The Stock Movement entity is the permanent log of changes to the quantity on hand. Each entry states the type of change, the number of units, and the balance that resulted. Entries are never edited or deleted. A mistaken entry is corrected by recording a new adjustment.

Notifications from other systems, such as a dispatch report from Delivery, carry a source event reference. Storing that reference prevents a repeated notification from deducting the same stock twice.

Table 5. Attributes of the Stock Movement Entity

Attribute	Key or Constraint	Description
movement_id	PK	Unique movement identifier
inventory_id	FK, required	Stock record that changed

Attribute	Key or Constraint	Description
reservation_id	FK, optional	Reservation fulfilled by this movement, for sales only
movement_type	Required	Restock, Sale, Return, Adjustment In, or Adjustment Out
quantity	Required, integer > 0	Units moved; the type determines whether stock increases or decreases
on_hand_after	Required, integer ≥ 0	Quantity on hand immediately after the movement
order_id	Optional	Related order for sales and returns
source_event_reference	Unique when supplied	Reference of the notification that caused the movement
performed_by	Required	User or connected system responsible for the change
remarks	Required for adjustments	Explanation of the change
created_at	Required	Date and time of the movement

2.3.4 Seller Offer — External Reference

Seller Offer represents one seller selling one product. Product/Catalog Management owns this entity. Inventory needs only its identifier and status to decide whether stock may be created or reserved. The attributes below follow the Catalog group's current draft and must be aligned with its final design.

Table 6. Referenced Attributes of Seller Offer

Referenced Attribute	Description
seller_id — PK, FK	Seller providing the offer
product_id — PK, FK	Product being offered
offer_status	Whether the offer is active or inactive

2.3.5 Order Item — External Reference

Order Item represents one seller offer within a customer order. Sales/Order Management owns this entity. Inventory uses its identifiers to link each reservation to the item it serves and reads the requested quantity from the reservation request.

Table 7. Referenced Attributes of Order Item

Referenced Attribute	Description
order_item_id — PK	Unique order item identifier
order_id — FK	Order that contains the item
offer_id — FK	Seller offer selected by the customer
quantity	Units requested by the customer

2.4 Relationships and Cardinalities

Table 8 states the cardinality and participation of each relationship shown in Figure 1.

Table 8. Relationships and Cardinalities of the Inventory Design

Relationship	Cardinality and participation
Seller Offer–Inventory	A seller offer may have zero or one stock record. Every stock record belongs to exactly one seller offer.

Relationship	Cardinality and participation
Inventory–Stock Reservation	A stock record may have zero or many reservations. Every reservation draws from exactly one stock record.
Order Item–Stock Reservation	An order item may be held by zero or one reservation. Every reservation serves exactly one order item.
Inventory–Stock Movement	A stock record may have zero or many movements. Every movement changes exactly one stock record.
Stock Reservation–Stock Movement	A reservation produces zero or one movement. A movement is linked to zero or one reservation.

3. BUSINESS RULES

3.1 Stock Record Rules

A stock record may be created only for a seller offer that exists in Product/Catalog Management. Each offer has at most one stock record. Quantities are whole numbers and may never fall below zero, and the reserved quantity may never exceed the quantity on hand.

The quantity on hand may change only through a recorded movement, and the movement must be saved in the same database transaction as the new balance. Stock records that already have movement history may not be deleted, because reservations and log entries depend on them. These requirements are listed in Table 9.

Table 9. Stock Record Business Rules

Rule	Requirement
BR-01	A stock record must reference an existing seller offer.
BR-02	A seller offer may have only one stock record.

Rule	Requirement
BR-03	Quantity on hand, quantity reserved, and reorder level must be whole numbers of zero or more.
BR-04	Quantity reserved may not exceed quantity on hand.
BR-05	Every change to quantity on hand must create a stock movement in the same transaction.
BR-06	Stock records with movement history may not be deleted.

3.2 Available Stock Calculation

The quantity that customers may still buy is computed from the two stored figures and is never saved as a separate value. Low-stock and out-of-stock conditions are derived from the same result.

Available Quantity = Quantity on Hand − Quantity Reserved

Low Stock: Available Quantity ≤ Reorder Level

Out of Stock: Available Quantity = 0

For example, a seller holds 15 wireless mice with a reorder level of 5. Two pending orders hold 4 and 3 units, so the available quantity is 15 − 7 = 8. If a third order reserves 3 more, the available quantity becomes 5 and the record is reported as low stock, even though 15 units are still physically present.

3.3 Reservation Rules

A reservation request from Sales lists every item of one order. Inventory checks all items before changing anything. If any item lacks enough available stock, or its seller offer is no longer active, the whole request is rejected and the response identifies the affected items. If every item passes, all reservations are created together.

Each stock record is locked while a reservation is being processed. This forces simultaneous requests for the same offer to be handled one after the other, which prevents two customers from receiving the same last unit. Table 10 lists the reservation rules.

Table 10. Reservation Business Rules

Rule	Requirement
BR-07	A reservation request is accepted only if every listed item can be reserved.
BR-08	New reservations require an active seller offer, product, and seller.
BR-09	Reserving increases quantity reserved only; quantity on hand is unchanged.
BR-10	A repeated request with the same request reference returns the original result.
BR-11	An order item may have only one reservation.
BR-12	An Active reservation expires automatically if payment is not confirmed before its expiry time.

3.4 Reservation Status Handling

The reservation status shows where a hold stands. It is separate from the order status kept by Sales and the shipment status kept by Delivery, although Inventory changes it in response to their results. Table 11 defines each status and its permitted next status.

Table 11. Reservation Statuses and Transitions

Status	Meaning	Normal next status
Active	Units are held and the reservation will expire if unpaid.	Confirmed, Released, or Expired
Confirmed	Payment is verified and the hold no longer expires.	Fulfilled or Released

Status	Meaning	Normal next status
Fulfilled	Items were dispatched and the units were deducted from stock.	None
Released	The hold was cancelled and the units returned to available stock.	None
Expired	The hold ended because payment was not confirmed in time.	None

Fulfilled, Released, and Expired are final. A Fulfilled reservation reduces both the quantity on hand and the quantity reserved and creates a Sale movement. A Released or Expired reservation reduces only the quantity reserved and creates no movement, since no unit left the seller.

3.5 Stock Movement Rules

Each movement type moves stock in one direction. Restock, Return, and Adjustment In increase the quantity on hand. Sale and Adjustment Out decrease it. Adjustments require remarks, and an Adjustment Out may not reduce the quantity on hand below the quantity reserved, because those units are already promised to customers.

Movements are never changed after they are saved. If a restock was entered with the wrong quantity, the seller records an adjustment for the difference, leaving both entries visible in the history.

3.6 Exception Handling

Table 12 lists the situations that fall outside the normal workflow and the response Inventory must apply in each case.

Table 12. Exception Handling Responses

Situation	Required response
Insufficient stock	Reject the whole reservation request and identify the items that are short.
Inactive seller offer	Reject new reservations; existing Confirmed reservations may still be fulfilled.
Simultaneous requests for the last unit	Lock the stock record so that only one request succeeds.
Repeated reservation request	Return the existing result found through the request reference.
Unpaid reservation past expiry	Mark the reservation Expired and return the units to available stock.
Order cancelled before dispatch	Release all active and confirmed reservations for the order.
Payment verified after expiry	Do not revive the reservation; Sales must submit a new request, which succeeds only if stock is still available.
Repeated dispatch notification	Ignore the event if its source reference was already recorded.
Returned package received	Record a Return movement and add the units back to stock.
Physical count differs from the record	Record an adjustment with remarks explaining the difference.

3.7 Access Control and Record Retention

Sellers may view their own stock records, record restocks, set reorder levels, and enter adjustments for their own offers. Authorized staff may review all records and enter

adjustments when required. Connected systems may request reservations, confirmations, releases, and deductions only through the agreed integration functions. No user may edit the reserved quantity directly.

Stock records, reservations, and movements are kept for review. Ordinary user actions may not delete them, since Sales, Delivery, and staff may need them to investigate an order.

4. INTEGRATION REQUIREMENTS

4.1 Integration Approach

Inventory exchanges information with Product/Catalog, Sales/Order, and Delivery/Shipment Management. It does not exchange information with Billing/Payment Management directly: payment results reach Inventory through Sales, which coordinates the order. This keeps one system responsible for deciding when an order moves forward.

Every request must state the identifiers Inventory needs and a unique reference, and every response must clearly report success, failure, or a pending result. A request that times out must not be treated as successful. The sender should repeat it with the same reference so that Inventory can return the original outcome.

4.2 Product/Catalog Management

Inventory receives seller offer identifiers and the status of each offer, product, and seller from Catalog. It uses them when a stock record is created and before any new reservation. In return, Inventory provides the available quantity and low-stock condition for each offer, which Catalog may display on product pages.

4.3 Sales/Order Management

Sales sends reservation requests containing the order identifier, each order item identifier with its seller offer and quantity, and a request reference. Inventory replies with

a single result for the order, the reservation identifiers, and the expiry time, or with the list of items that could not be reserved.

Sales also notifies Inventory when payment is verified, so that the reservations become Confirmed, and when an order is cancelled or payment fails, so that the reservations are released. The groups must agree on how the offer identifier used by Sales maps to the seller and product pair used by Inventory.

4.4 Billing/Payment Management

No direct exchange is proposed. Billing reports payment results to Sales, and Sales forwards the effect on stock to Inventory. If the class later decides that Billing should notify Inventory directly, the same confirmation and release functions can be reused without changing the database design.

4.5 Delivery/Shipment Management

Delivery may read the reservation status of an order to confirm that its stock is still held before preparing a shipment. When the package is dispatched, Delivery sends the order identifier, the order item identifiers, and a source event reference. Inventory then marks the reservations Fulfilled, deducts the units, and records Sale movements.

When a failed or returned delivery reaches the seller, Delivery sends a return notification with its own event reference. Inventory records Return movements and adds the units back to the quantity on hand.

4.6 Integration Summary

Table 13 consolidates the information exchanged between Inventory and each connected system.

Table 13. Integration Summary of the Inventory Management System

Connected system	Inventory receives	Inventory provides
Product/Catalog	Seller offer identifiers and the status of offers, products, and sellers	Available quantity and low-stock condition per offer
Sales/Order	Reservation requests, payment confirmation, cancellation, and failed-payment notices	Reservation results, reservation identifiers, expiry times, short-item lists, and release confirmations
Billing/Payment	None directly; payment results arrive through Sales	None directly
Delivery/Shipment	Dispatch and return notifications with event references	Reservation status per order and deduction confirmations

4.7 Shared Processing Requirements

All groups must use the same identifier formats, status names, and timestamp conventions. Inventory requires a unique reference on every request and event it receives, and it rejects updates that do not fit the current reservation status, such as a dispatch notice for a released reservation.

When a result is missing or contradictory, the affected reservation stays in its current status until the responsible systems resolve the difference. Inventory never guesses a result in order to move a reservation forward.

5. PROPOSED END-TO-END WORKFLOW

The process begins when a seller records stock for an active seller offer. Inventory creates the stock record and logs a Restock movement. From that point, the offer's available quantity can be shown to customers.

When a customer submits an order, Sales sends a reservation request. Inventory checks every item, locks the affected stock records, and either reserves all items with a 30-minute expiry or rejects the request with the list of short items. The quantity on hand does not change at this stage.

After Billing verifies payment, Sales notifies Inventory and the reservations become Confirmed. If payment fails or the customer cancels, Sales requests a release. If no notice arrives before the expiry time, Inventory expires the reservations on its own. In each of these cases, the units return to available stock.

When Delivery dispatches the package, Inventory marks the reservations Fulfilled, reduces the quantity on hand and the quantity reserved, and records Sale movements. If the package later returns to the seller, Inventory records Return movements. Throughout the process, every change to the physical count leaves a permanent entry in the movement log.